Quiz 6

1. Polyphase definition

For $M = 3$, write the definition of the k-th polyphase sequence $e_k[r]$ in terms of $h[n]$.

$e_k[r] = h[rM + k], k = 0,1, \dots, M - 1.$

2. Arithmetic speedup

A 30-tap FIR is followed $\downarrow 5$. Approximately, what ideal arithmetic speedup does a polyphase decimator provide relative to filtering every full-rate output?

Approximately $5\times$.

3. Noble identity

True or false: $H(z) \rightarrow \downarrow M$ is always equivalent to $\downarrow M \rightarrow H(z)$. Explain.

False. A filter may commute across $\downarrow M$ in the Noble identity when it has the special $G(z^M)$ form before the downsampler; arbitrary $H(z)$ does not commute.

4. M=2 decomposition

For $h = \{1, 2, 3, 4, 5, 6\}$ and $M = 2$ find $e_0[r]$ and $e_1[r]$.

$e_0 = \{1,3,5\}$, $e_1 = \{2,4,6\}$.

5. Interpolation work

A 24-tap FIR interpolates by $L = 3$. Compare the approximate naïve and polyphase multiply rates in units of F_s.

Naïve $\approx 24 \cdot 3F_s = 72F_s$ multiplies/s; polyphase $\approx 24F_s$; about $3\times$ saving.

6. Perfect reconstruction

In a two-channel perfect-reconstruction bank, what two broad requirements must be satisfied?

Alias terms must cancel (“alias term”). The desired transfer path is only scale and delay (“distortion term”, no frequency-dependent distortion).